



Contact

Discord Username
drlove.bot

Email
sudiproy20yo@gmail.com

Github Account
<https://github.com/drlove2002>

LinkedIn
www.linkedin.com/in/drlove2002

Education

2020 - 2023
B.Sc. in Computer Science
University of Calcutta

Expertise

- Backend Development (Python, Rust, gRPC)
- Microservices Architecture
- Databases (MongoDB, PostgreSQL, libSQL)
- Caching Strategies (Redis, Embedded SQLite)
- API Development (Axum, Actix)
- Web Development (Next.js, TypeScript, HTML, CSS, JQuery, Node.js)
- DevOps (CI/CD, GitHub Actions, Docker, AWS)
- Statistical Modeling (5-gram algorithm)
- Performance Optimization
- Scalable System Design

Language

- Bengali (Native)
- English (Fluent)
- Hindi (Intermediate)

Sudip Roy

Full Stack Developer

Full Stack Developer with a Bachelor's degree in Computer Science, specializing in back-end and software engineering. Skilled in Python, Rust, TypeScript, CI/CD, system design, GitHub, Docker, and microservice architecture. Proven track record of delivering cutting-edge solutions to complex problems. Intermediate business knowledge with practical experience in the stock market. Known for intellectual curiosity, problem-solving abilities, and a unique perspective that enhances team dynamics.

Experience

2023

Final Year Project Report

Interactive Game with Statistical Model

- Led the development of an engaging game, achieving a 20% increase in user interaction by predicting inputs with a 5-gram algorithm.
- Implemented strategies to encourage users to defy predictability, fostering a 15% increase in diverse decision-making.
- Engineered backend using Rust, Actix, and Redis, enhancing data handling efficiency by 30%.
- Developed frontend with TypeScript, HTML, JQuery, Webpack, SASS, and Node.js, boosting interactivity by 25%.
- Implemented CI/CD pipelines with GitHub Actions and Docker, scaling deployment on AWS by 50%.
- Demonstrated the interplay between human behavior and statistical prediction models through interactive visualizations.

2020 - Present

Worldwide Discord Server

Software Developer (Discord Bot)

- Developed and maintained a Python-based Discord bot, increasing server activity and member retention.
- Implemented an economy system, boosting member engagement by 40% through virtual currency rewards.
- Utilized data storage solutions, improving data retrieval speed with JSON files, MongoDB, Redis, PostgreSQL, and SQLite.
- Leveraged Redis as a caching layer, enhancing performance by 20% and reducing database load by 30%.
- Migrated to a Rust-based Data Access Layer (DAL) using gRPC, improving interservice communication efficiency by 25%.
- Integrated webhooks and APIs (e.g., Axum), enabling virtual currency purchases through real-money donations, increasing revenue by 15%.
- Optimized performance by 30% using an embedded SQL database (libSQL fork of SQLite) within the Rust process.
- Planned and developed a Next.js web dashboard hosted on AWS and Cloudflare CDN for bot management streamlining bot management processes by 40%.
- Utilized hosting platforms like repl.it, DigitalOcean, and AWS EC2, improving deployment and scaling efficiency by 50%.